

from (i) & (ii)

$$E 4\pi r^2 = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

$$E = \frac{\sigma}{4\pi\epsilon_0 r^2}$$

$$E \propto \frac{1}{r^2}$$

for outside.

(ii) field at surface of shell.

$$E = \frac{\sigma}{4\pi\epsilon_0 R^2}$$

$$r = R$$

(iii) field inside the shell.

the point P inside the shell.

Here Gaussian Surface encloses no charge.

$$\phi = E \Delta s$$

$$\phi = 4\pi r^2$$

$$\phi = E 4\pi r^2 \quad \text{--- (i)}$$

$$\phi = \frac{q_{en}}{\epsilon_0}$$

$$\phi = \frac{0}{\epsilon_0} = 0 \quad \text{--- (ii)}$$

$$E = 0$$

